

Chapter I

Introduction to Virology

Learning objectives

Upon completion of this chapter, the student will be able to:

- 1. Define virus.**
- 2. Describe structure & properties of medically important viruses**
- 3. Illustrate classification of medically important viruses**
- 4. Explain virus infectious cycle**
- 5. Describe antiviral agents and their mechanism of action**

Chapter I

Introduction to Virology

- **Virology** is the discipline of **microbiology** that is concerned with the study of **viruses**.
 - Viruses are autonomous infectious microorganismes
 - They have no metabolic systems of their own
 - The smallest infectious agents (Size: 20 – 400 nm in diameter)
 - Are obligatory intracellular parasites of
 - bacteria, protozoa, fungi, algae, plants, and animals
 - Cannot make energy or proteins independent of a host cell
 - Viral genomes may be RNA or DNA , but not both

- Viruses consist of **nucleic acid** surrounded by a protein **coat**.
- Some viruses have additional lipoprotein **envelope**.
- Viruses **lack** cellular organelles, such as **mitochondria** and **ribosomes**.
- They replicate only inside living cells.
- **Viroids** (The complete virus particle) A fully assembled infectious virus, nucleic acid and capsid
- **Prions**

Infectious particles that are **entirely protein**, No nucleic acid and Highly heat **resistant**

Affects nervous tissue and results in Bovine spongiform encephalitis (BSE) “mad cow disease”,
Scrapie in sheep kuru & Creutzfeldt-Jakob Disease (CJD) in humans

Comparison of bacteria, rickettsiae, chlamydiae, and viruses

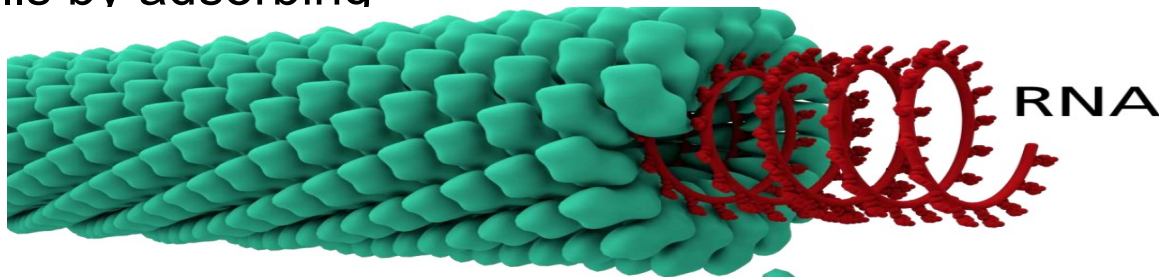
Character	Typical bacteria	Rickettsiae	Chlamydiae	Viruses
Intracellular parasite	No	Yes	Yes	Yes
Cellular organization	Yes	Yes	Yes	No
Plasma membrane	Yes	Yes	Yes	No
Replication by binary fission	Yes	Yes	Yes	No
Growth on inanimate media	Yes	No	No	No
Pass through bacteriological filters	No	No	Yes	Yes
Possess both DNA and RNA	Yes	Yes	Yes	No
ATP-generating metabolism	Yes	Yes	No	No
Ribosome	Yes	Yes	Yes	No
Sensitive to antibiotics	Yes	Yes	Yes	No
Sensitive to interferon	No	No	Yes	Yes

Structural Components of Viruses

Capsid

- Viral shell which encloses viral nucleic acid, protein in nature
- Composed of a large number of protein subunits (polypeptides) called ***capsomere***
 - a single type in some viruses, while in other viruses several types
- The capsid **protects** nucleic acid and **facilitates entry** of viral genome into the host

cells by adsorbing



Viral envelopes

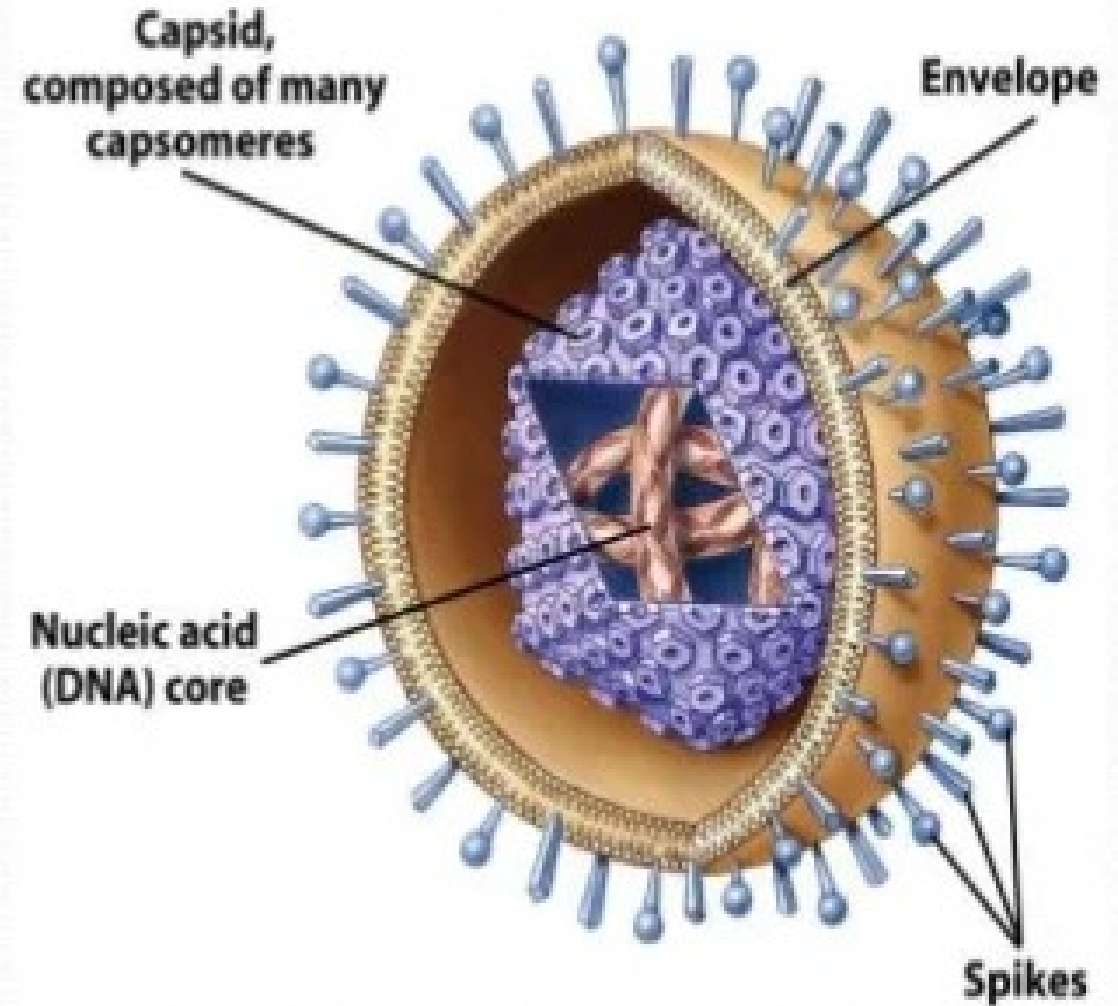
- An outer membrane located outside the capsid
- Made up of **lipid** and **carbohydrate** of **host** cell membranes
- It is acquired during viral maturation
- Different in different viruses so can be used for the **identification**
- Flexible so show variable shapes (also called **pleomorphic**)
- Contain virus encoded **glycoproteins**
- May or may not be covered by spikes
- Lipid enveloped viruses are **sensitive** to **Ether** and other organic solvents ► loss of infectivity

Genome

- Nucleic acid DNA or RNA
- Shape Linear, Circular, Segmented
- Strandedness Single stranded, Double- stranded
- Sense Positive sensed (+)
 Negative sense (-)
- **Size variable**
 - 3.2kb (Hepadnavirus) – 375 kb DNA viruses
 - 7 kb (picornavirus) - 30 kb (Corona virus)

Viral components

- Nucleic acids
- Capsid
- Envelope



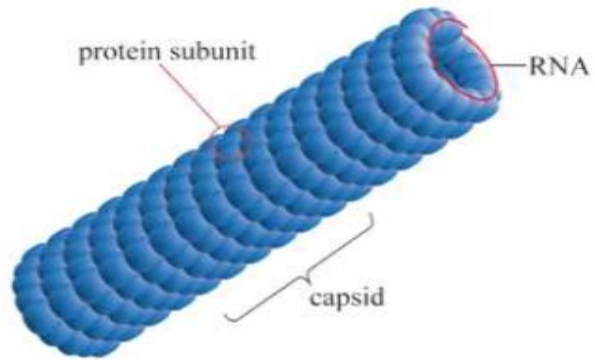
Schematic diagram of structure of a virus

Symmetry of viruses

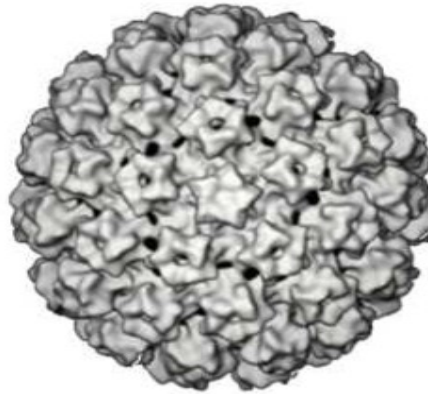
- On the basis of capsid structure, the viruses can be classified into **three** types of symmetry
- **Polyhedral viruses (Cubic symmetry)**
 - appear as many sided viruses
 - consist of capsids in the shape of an icosahedron
 - The number and arrangement of the capsomeres, identification and classification
- **Helical symmetry:**
 - The virus particle is elongated or pleomorphic (not spherical), and the nucleic acid is spiral.
 - Capsomeres are arranged round the nucleic acid
- **Complex symmetry:**
 - The virus particle does not confirm either cubic or helical symmetry.

Symmetry of virus

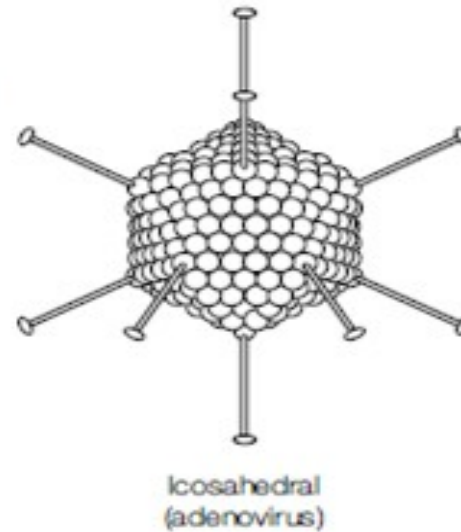
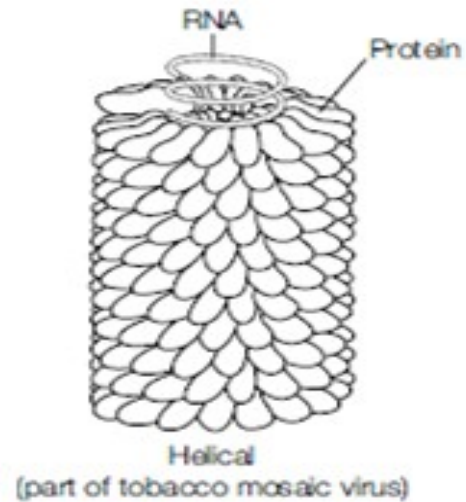
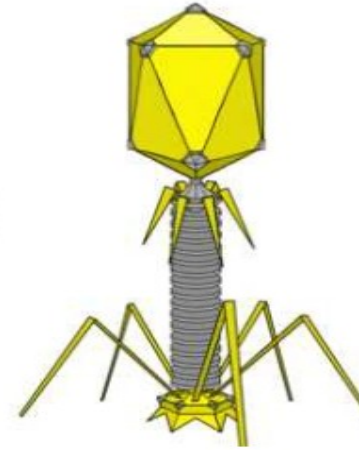
Helical



Icosahedral



Complex



Susceptibility to Physical and Chemical Agents

- **Disinfectants:** resistant than bacteria to chemical disinfectants
 - hydrogen peroxide, potassium permanganate, hypochlorite, formaldehyde and beta-propiolactone most active agents
- **Temperature:** Most of the viruses are highly heat labile inactivated within seconds at 56°C
 - viruses are stable at low temperature
- **pH:** remain viable in a pH range of 5–9
 - sensitive to extremes of acidity and alkalinity
- **Lipid solvents**
 - Ether, chloroform, and detergents are active against enveloped viruses
- **Radiations:** readily inactivated by sunlight, ultraviolet (UV) radiations, and ionizing radiations

Classification of viruses

1. Type of Nucleic Acid

- DNA or RNA (Ss /Ds)
- Strategy of replication

2. Size & morphology

- Type of symmetry (Icosahedral, Helical . Complex)
- Number of capsomers
- Presence or absence of envelope

3. Presence of specific enzymes

- RNA and DNA polymerase
- Neuraminidase
- Reverse transcriptase

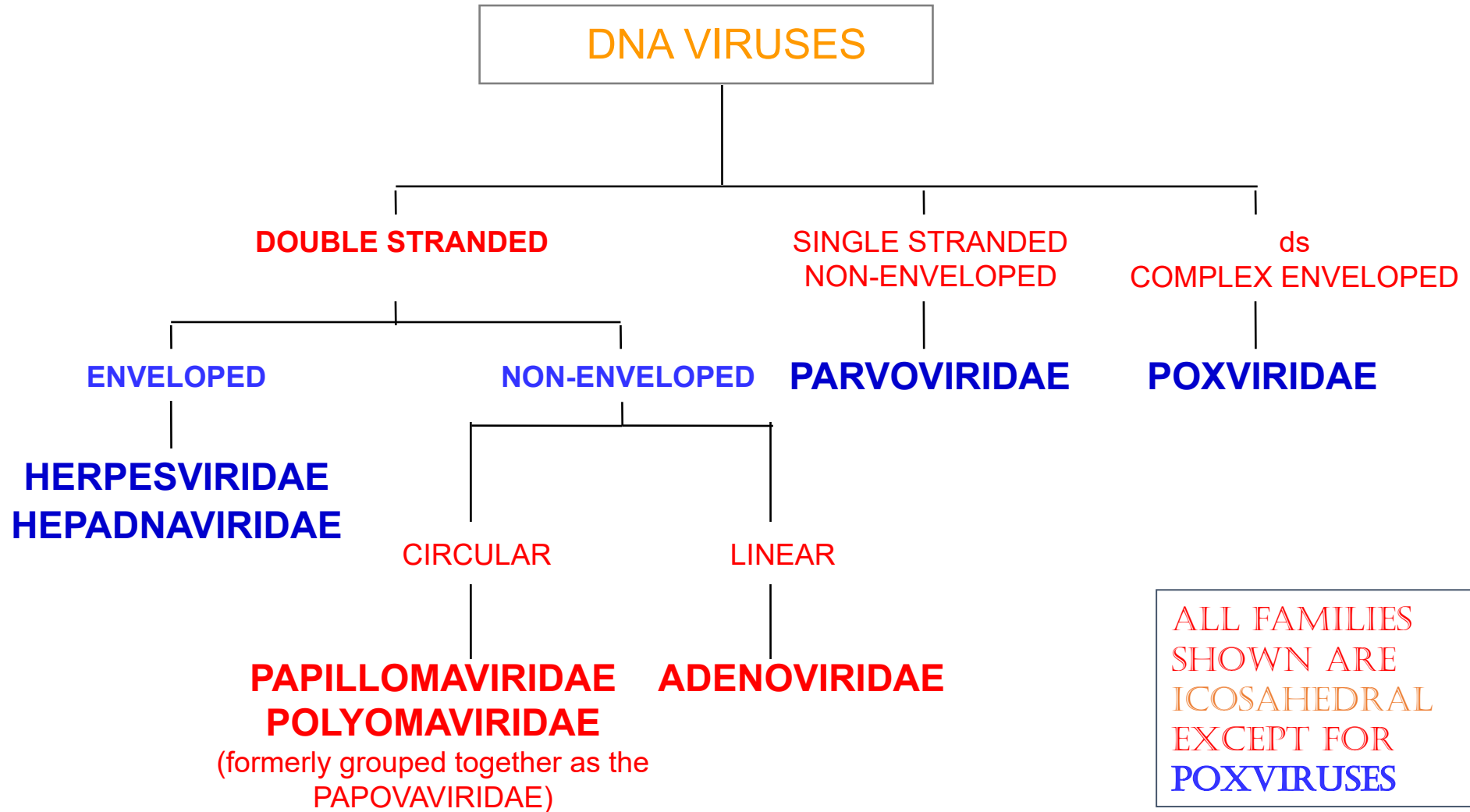
4. Host tissue or cell tropism

- Hepatitis viruses, HIV, etc

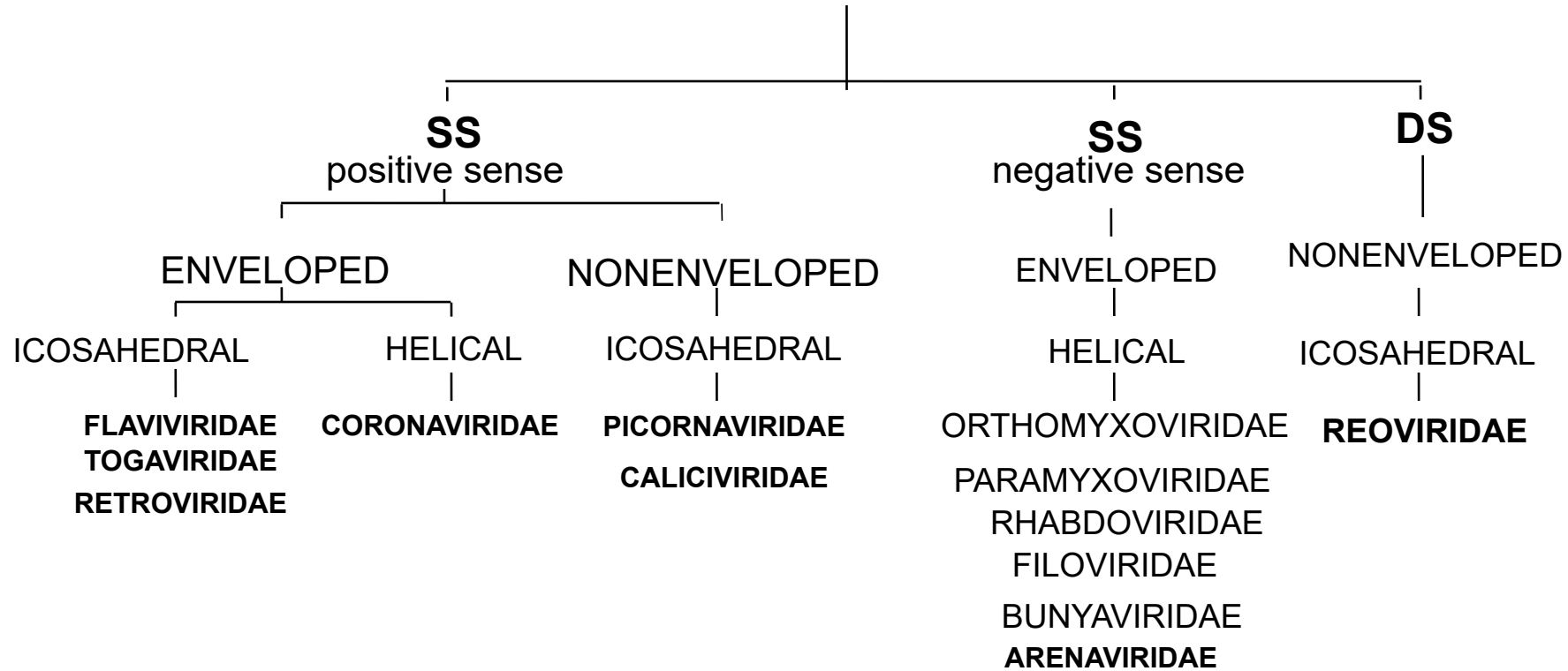
5. Mode of transmission e.g. Arboviruses

6. Host range E.g. Animal, bacteria, plant

7. Type of disease E.g. encephalitis



RNA VIRUSES



Classification on the basis of replication properties and site of replication

➤ **Replication and assembly in cytoplasm of host:**

- All RNA virus except Influenza virus

➤ **Replication in nucleus and assembly in cytoplasm of host:**

- Influenza virus, Pox virus

➤ **Replication and assembly in nucleus of host:**

- All DNA viruses except Pox virus.

➤ **Virus replication through ds DNA intermediate:**

- All DNA virus, Retro virus and some tumor causing RNA virus

➤ **Virus replication through ss RNA intermediate:**

- All RNA virus except Reo virus and tumor causing RNA viruses.

Classification on the basis of host range:

- **Bacteriophage:** Phage are virus infecting bacteria.
- **Plant virus:** Those virus that infects plants. Eg. TMV, cauliflower mosaic virus
- **Animal virus:** Those virus that infects animals. Eg. Polio virus, Retro virus, Herpes virus, Adeno virus
- **Insect virus:** virus that infects insects. Eg. Baculovirus, Sacbrood virus, Entomopox virus, Granulosis virus

Classification of Human Viruses

<u>"Group"</u>	<u>Family</u>	<u>Genome</u>	<u>Genome size (kb)</u>	<u>Capsid</u>	<u>Envelope</u>
dsDNA	Poxviridae	dsDNA, linear	130 to 375	Ovoid	Yes
	Herpesviridae	dsDNA, linear	125 to 240	Icosahedral	Yes
	Adenoviridae	dsDNA, linear	26 to 45	Icosahedral	No
	Polyomaviridae	dsDNA, circular	5	Icosahedral	No
	Papillomaviridae	dsDNA, circular	7 to 8	Icosahedral	No
ssDNA	Anellovirus	ssDNA circular	3 to 4	Isometric	No
	Parvoviridae	ssDNA, linear, (- or +/-)	5	Icosahedral	No
Retro	Hepadnaviridae	dsDNA (partial), circular	3 to 4	Icosahedral	Yes
	Retroviridae	ssRNA (+), diploid	7 to 13	Spherical, rod or cone shaped	Yes
dsRNA	Reoviridae	dsRNA, segmented	19 to 32	Icosahedral	No
ssRNA (-)	Rhabdoviridae	ssRNA (-)	11 to 15	Helical	Yes
	Filoviridae	ssRNA (-)	19	Helical	Yes
	Paramyxoviridae	ssRNA (-)	10 to 15	Helical	Yes
	Orthomyxoviridae	ssRNA (-), segmented	10 to 13.6	Helical	Yes
	Bunyaviridae	ssRNA (-, ambi), segmented	11 to 19	Helical	Yes
	Arenaviridae	ssRNA (-, ambi), segmented	11	Circular, nucleosomal	Yes
	Deltavirus	ssRNA (-) circular	2	Spherical	Yes
ssRNA (+)	Picornaviridae	ssRNA (+)	7 to 9	Icosahedral	No
	Calciviridae	ssRNA (+)	7 to 8	Icosahedral	No
	Hepevirus	ssRNA (+)	7	Icosahedral	No
	Astroviridae	ssRNA (+)	6 to 7	Isometric	No
	Coronaviridae	ssRNA (+)	28 to 31	Helical	Yes
	Flaviviridae	ssRNA (+)	10 to 12	Spherical	Yes
	Togaviridae	ssRNA (+)	11 to 12	Icosahedral	Yes

Baltimore Classification

- Distinguishes between viruses whose **genomes** can be utilized **directly as mRNA** vs those that require a virion-associated **enzymes** to produce mRNAs (negative stranded RNA and dsRNA viruses)
 - Positive-sense RNA viruses vs other RNA viruses
 - All RNA viruses must encode their own polymerase because cells do not have RNA-dependent RNA polymerase activity

Pathogenesis of viral infection

- Viruses cause infection of the host:
 1. Breaking the natural protective mechanisms
 2. Evading the immune system
 3. Killing off the host cells and triggering immune and inflammatory responses

Viral Pathogenesis

- Entry
- Primary replication
- Cell tissue tropism
- Spread throughout the host
- Host immune response
- Secondary replication
- Cell / tissue damage
- Persistence / clearance

Portals of Entry for Viruses

- Respiratory tract (droplets, air)
 - Influenza viruses, RSV, EBV, HSV, CMV
 - measles, mumps & rubella virus, Varicella-Zoster-Virus
- Gastrointestinal tract (faecal-oral)
 - rota-, adeno-, calici- and enteroviruses, Hepatitis A Virus
- Blood & blood products (parenteral)
 - HIV, Hepatitis B and C Viruses
- Bites and stings (vertebrates/insects)
 - Arbo virus, rabies virus

Virus infectious cycle

- The multiplication of viruses, both DNA and RNA viruses, is divided into six phases
- Attachment
 - Penetration
 - Uncoating
 - Biosynthesis
 - Maturation
 - Release

1. Adsorption or attachment

- Reactive sites on viruses surface interact with specific receptors on susceptible host cells
 - hemagglutinin of influenza virus
 - some complex viruses (HSV) may have more than one -receptor molecule
 - HIV – CD4 + T cells
- Receptors on the virus capsid or envelope **irreversibly** binds to cellular receptors on the cell membrane
- Receptor molecules differ for different viruses but are generally glycoproteins.

2. **Penetration:** an energy-dependent step involves one of three mechanisms

- **translocation** of the virion across the plasma membrane,
- **endocytosis** of the virus particle resulting in accumulation of virions inside cytoplasmic vacuoles and
- **fusion** of the cellular membrane with the virion envelope
 - Non-enveloped viruses penetrate by the first two mechanisms

3. **Uncoating**

- Viral capsid is broken by viral or cellular enzyme (lysosomal);
- The infectivity of the virus is lost at this point.
- Uncoating is followed by the "eclipse phase", when no virions are seen.
- Viral NA is released;

4. Biosynthesis or genomic activation

The sites of viral synthesis depend on the type of virus, RNA virus or DNA virus

➤ DNA viruses:

- DNA viruses replicate their DNA in the nucleus of the host cell by using **viral enzymes**
- synthesize their capsid and other proteins in the cytoplasm by using **host cell enzymes**
- the proteins enter into the nucleus and assemble with the newly synthesized DNA to form virions
- these virions are transported along with **endoplasmic reticulum** to the membrane of the host cell for release.

➤ RNA viruses:

- multiply in the cytoplasm of the host cell.
- Varies among the species based on how mRNA and viral RNA are produced
 - double-stranded RNA is transcribed to mRNA by viral polymerase
 - Single-stranded RNA viruses
 - Positive sense RNA virus, the viral RNA itself acts as the mRNA
 - Negative sense RNA viruses possess their own **RNA polymerase** for transcription of mRNA
 - single-stranded RNA genome
 - Convert their genome to **cDNA** by **reverse transcriptase** enzyme
 - mRNA is generated from the cDNA

5. Assembly or Maturation

- The assembly of the protein capsid is the first step in viral maturation
- During maturation the **envelope protein** is encoded by the **viral genes** and is incorporated into the plasma membrane of the infected host cell
- The **envelope lipids** and **carbohydrates** are encoded by **host cell genes**, and are present in the plasma membrane
- The envelope develops around the capsid by a process called ***budding***
- The non-enveloped viruses are present intracellularly as fully developed viruses

N.B: Accumulation of virions at sites of assembly may form inclusion bodies

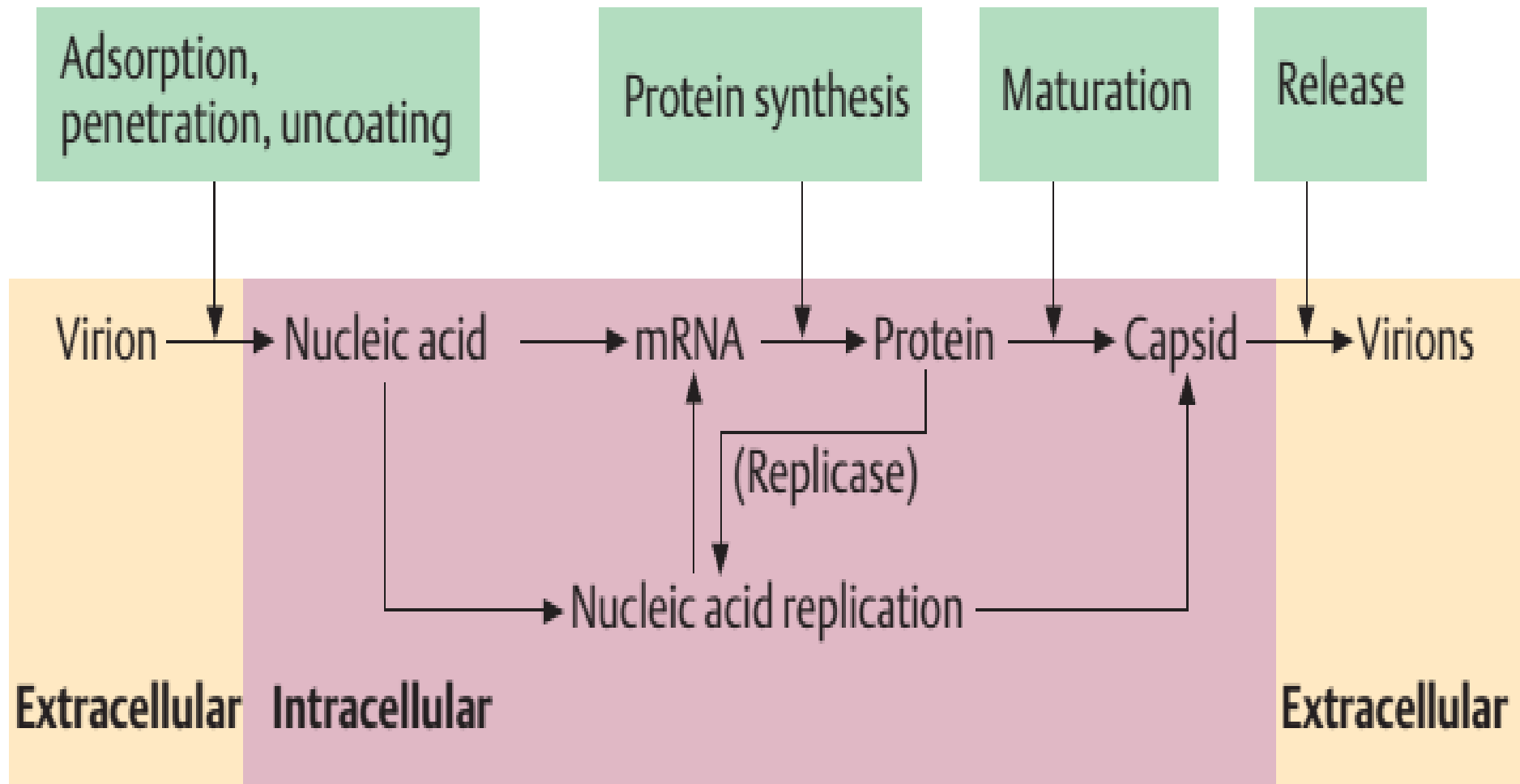
► **Visible in stained cells with light microscope**

6. Release

Release of the completed viruses is the last step in the replication of viruses

- Release of new intact infectious virions
- May occur by:
 - Budding E.g: Enveloped viruses
 - Lysis of infected host cells/ tissues

Virus Replication



Abnormal Replicative Cycles

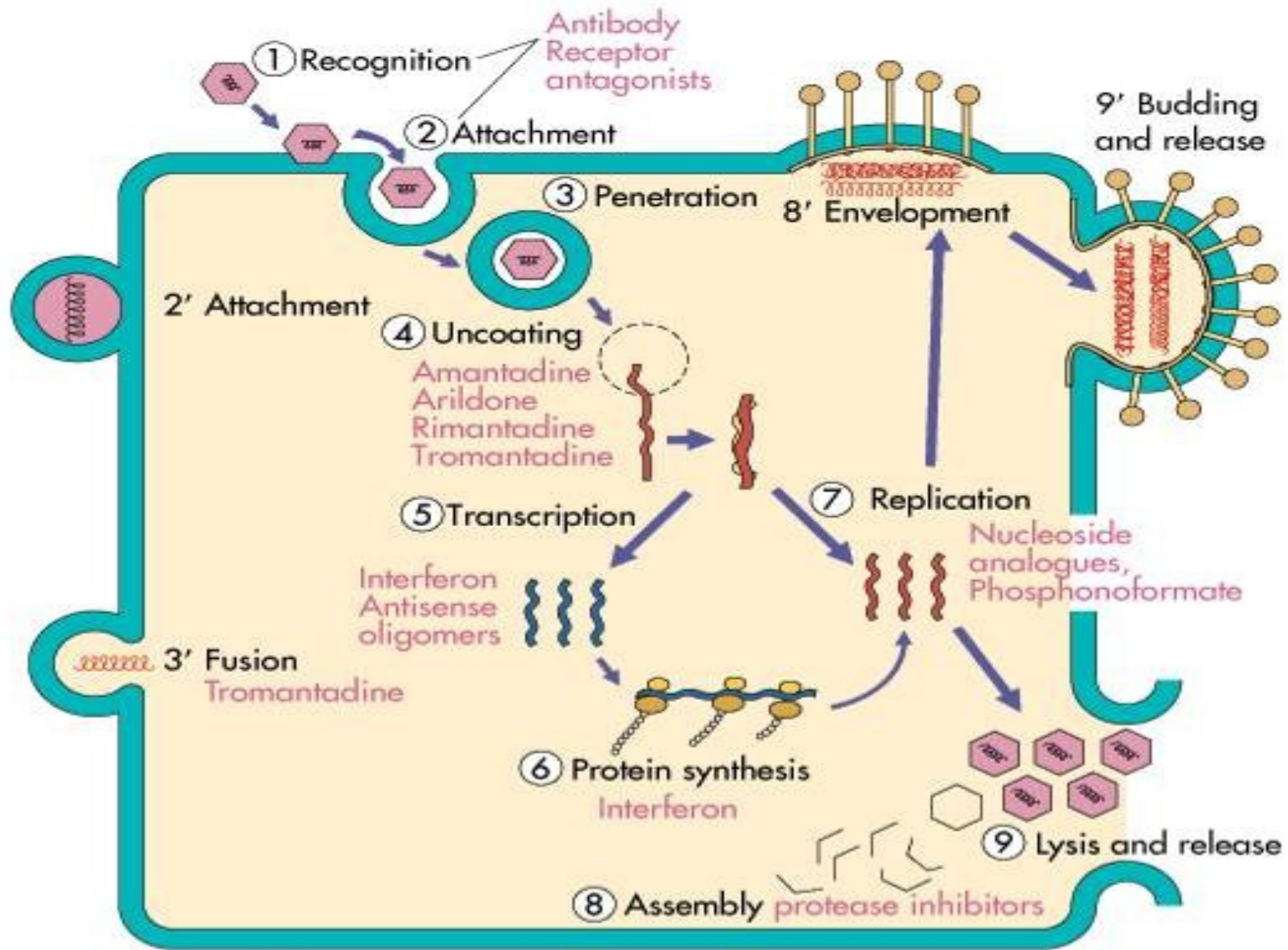
- During replicative cycles of virus, many types of abnormal replicative cycle do occur
 - **Incomplete viruses:** Due to defect during assembly of viral components the daughter virions may not be infective
 - **Pseudovirions:** occasionally enclose host cell nucleic acid instead of viral nucleic acid
 - **Abortive infections:** the virus components may be synthesized but the maturation is defective
 - **Defective viruses:** viruses that are genetically defective do not produce fully formed virions.

- Many viruses encode virulence factors that promote the efficiency of
 - viral replication,
 - viral transmission,
 - access and binding of the virus to target tissue, or
 - escape of the virus from host defense and immune resolution.
- Loss of virulence factors results in attenuation of the virus.
- Viruses also encode enzymes that are not present in un-infected cell.
 - critical to viral replication but unnecessary for cellular function.
- It is now possible to find specific inhibitors for viral growth.

❑ The greatest success in antiviral chemotherapy has been achieved by using:

- inhibitors of nucleic acid synthesis.
- The stages of **attachment** of virus to host cell
- Stage of **uncoating** of the viral genome
- Reverse transcription of certain viral genomes
- Regulation of viral transcription
- Replication of viral nucleic acid
- Translation of viral proteins
- **Assembly**
- Maturation
- **Release** of progeny virus

The mechanism of action varies among anti-virals.



Other major targets:
 Nucleotide biosynthesis and mutation: ribavirin
 Thymidine kinase (drug activation): acyclovir, penciclovir
 Neuraminidase: zanamivir, oseltamivir

Antiviral agents and their targets

Drug:	Viruses	Chemical Type	Target
<u>Vidarabine</u>	<u>Herpesviruses</u>	Nucleoside analogue	Virus polymerase
Acyclovir	Herpes simplex (HSV)	Nucleoside analogue	Virus polymerase
<u>Gancyclovir</u>	Cytomegalovirus (CMV)	Nucleoside analogue	Virus polymerase (needs virus UL98 kinase for activation)
<u>Nucleoside-analog reverse transcriptase inhibitors (NRTI):</u> <u>AZT (Zidovudine),</u> <u>ddI (Didanosine),</u> <u>ddc (Zalcitabine),</u> <u>d4T (Stavudine),</u> <u>3TC (Lamivudine)</u>	Retroviruses (HIV)	Nucleoside analogue	Reverse transcriptase

Antiviral agents

<u>Non-nucleoside reverse transcriptase inhibitors (NNRTI):</u> <u>Nevirapine,</u> <u>Delavirdine</u>	Retroviruses (HIV)	Nucleoside analogue	Reverse transcriptase
<u>Protease Inhibitors:</u> Saquinavir, Ritonavir, Indinavir, Nelfinavir	HIV	Peptide analogue	HIV protease
<u>Ribavirin</u>	Broad spectrum: HCV, HSV, measles, mumps, Lassa fever	<u>Triazole carboxamide</u>	<u>Virus replicase</u> (complex action, direct or indirect)
<u>Amantadine / Rimantadine</u>	Influenza A strains	<u>Tricyclic amine</u>	Matrix protein / <u>haemagglutinin</u>
<u>Interferons</u>	Hepatitis B and C	Protein	Cell defense proteins activated

Viral vaccines

- A vaccine is a biological preparation that provides
 - active acquired immunity
- A vaccine typically contains an agent that resembles a **disease-causing microorganism**
- Purpose of viral vaccines is to utilize immune response of the host to prevent viral disease
- Several remarkably effect
 - Small pox eradicated
 - Reducing annual incidence several viral diseases

There are several types of vaccines, including:

- **Inactivated vaccines**

- use the killed version of the germ that causes a disease.
- Inactivated vaccines are used to protect against: [Hepatitis A](#), [Flu](#), [Polio](#) and [Rabies](#)

- **Live-attenuated vaccines**

- use a weakened (or attenuated) form of the germ that causes a disease
- Live vaccines are used to protect against: [Measles](#), [mumps](#), [rubella](#) (MMR combined vaccine), [Rotavirus](#), [Smallpox](#), [Chickenpox](#) and [Yellow fever](#)

- **Subunit, recombinant, polysaccharide, and conjugate vaccines**

- use specific pieces of the germ—like its protein, sugar, or capsid
- need booster shots to get ongoing protection against diseases
- These vaccines are used to protect against: [Hepatitis B](#), [HPV](#)

Most Common Viral Vaccines

Vaccine

Type of Vaccine

Measles

Live, attenuated

Mumps

Live, attenuated

Rubella

Live, attenuated

Varicella zoster

Live, attenuated

Polio

Killed, intramuscular

Polio

Live, attenuated, oral

Influenza

Inactivated

Influenza

Live, attenuated, aerosol

Hepatitis A

Inactivated

Hepatitis B

Virus-like particle

Rabies

Inactivated

Rotavirus

Live, attenuated reassortant, oral

Human papillomavirus

Virus-like particle